

Systems Programming Roles and Facilities



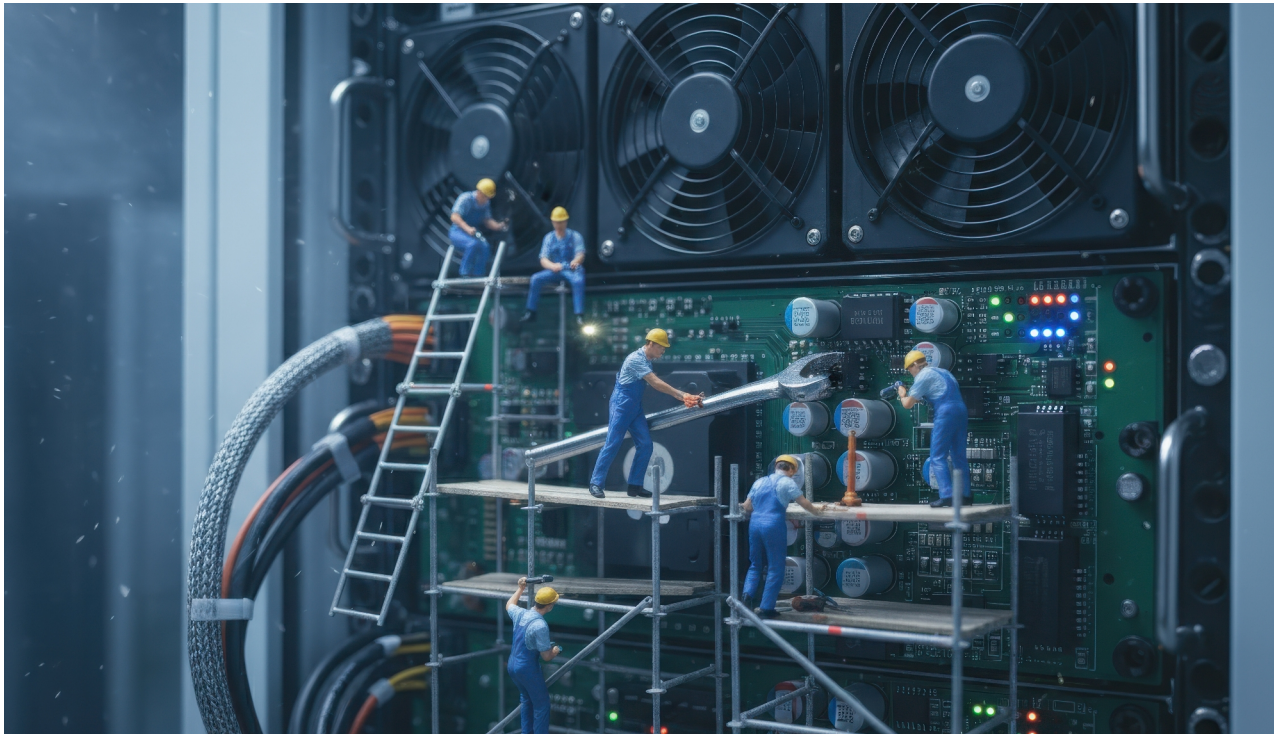


TITLES AND ROLES

Similar and overlapping roles and responsibilities.

Terms originate from different professional communities.

Scope of work for the same title varies from company to company, depending on their definition of *system*, *platform*, and *infrastructure*.



System Administrator – takes care of day-to-day operational and configuration details, installs and uninstalls software

Systems Programmer (1) – OS internals specialist, writes custom low-level code

Systems Programmer (2) – same as System Administrator, but for mainframes

Platform Engineer (1) – designs computer systems from the ground up

Platform Engineer (2) – builds and supports application delivery pipelines and automated systems to support application development

Platform Engineer (3) – same as System Administrator

Infrastructure Engineer (1) – designs and builds network infrastructures

Infrastructure Engineer (2) – same as Platform Engineer (2)

TYPICAL RESPONSIBILITIES FOR Z/OS

- Install, update, uninstall z/OS, optional OS components, and third-party software
- Configure z/OS and other software
- Manage system libraries
- Configure system security
- Ensure system reliability and performance
- Troubleshoot issues
- Assist application developers with JCL, system utilities, and configuration settings
- Build and support delivery pipelines
- Assist with automation requirements



WHAT MIGHT YOU DO AS A PLATFORM ENGINEER?



- Configure z/OS to auto-start DB2 during IPL
- Write shell scripts to run under USS to automate system administration tasks
- Optimize SQL queries for DB2 applications
- Customize the ISPF Primary Option Menu
- Help an application team set up a CI/CD pipeline
- Modify security settings for a user account
- Test a new release of CICS to ensure it is safe to put into production
- Brew coffee in the company break room
- Monitor the production environment to ensure critical systems are active and reachable
- Write a program to display customer account statuses and balances
- Use Ansible to automate sysadmin tasks across Linux, Windows, and z/OS instances
- Assist a developer who is having difficulty with JCL

Z/OS TOOLS & FACILITIES

Install, update, uninstall z/OS, optional OS components, and third-party software	z/OSMF, SMP/E, TSO/ISPF, JCL, USS, Ansible
Configure z/OS and other software	TSO/ISPF, JCL, USS, Ansible
Manage system libraries	TSO/ISPF, JCL, DFSMS, Ansible
Configure system security	RACF
Ensure system reliability and performance	RMF, SMF, others
Troubleshoot issues	Various
Assist application developers with JCL, system utilities, and configuration settings	Various
Build and support delivery pipelines	TSO/ISPF, JCL, USS, Git, Version Control, Ansible
Assist with automation requirements	TSO/ISPF, JCL, USS, REXX, others



WHAT YOU KNOW AND WHAT YOU CAN DO

- You understand the kinds of tasks that might be relevant to a role like Platform Engineer.
- You have a general idea of which z/OS tools and facilities support each kind of task.
- You know there is some variability in the meaning of terms and the expectations of people in particular roles.

WHAT'S NEXT

- TSO and ISPF